

Final HW

5.1.1 Results

1. Script used for hand calculation.

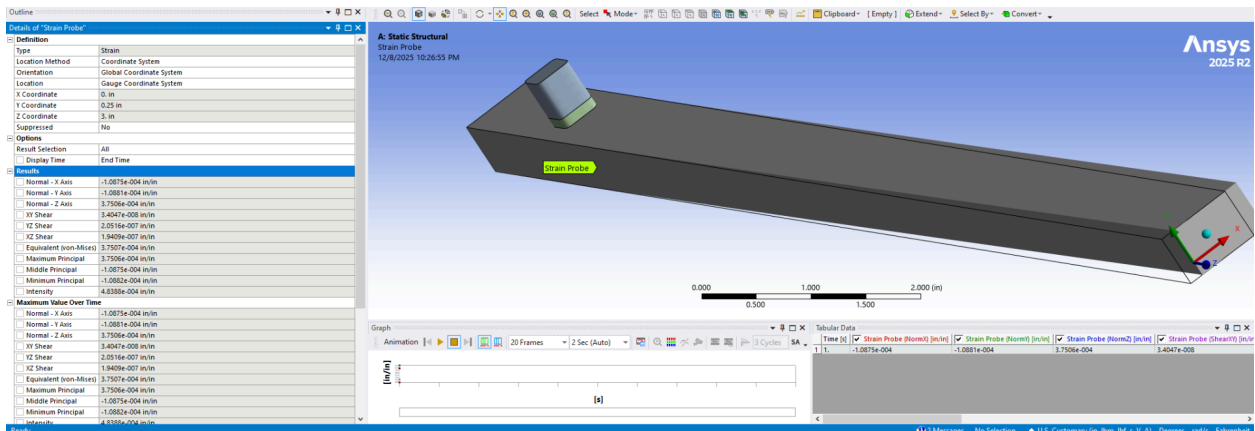
```
1 % Baseline Design Parameters
2 M = 600; % max torque (in-lbf)
3 L = 16; % length from drive to where load applied (inches)
4 h = 0.75; % width
5 b = 0.5; % thickness
6 c = 1.0; % distance from center of drive to center of strain gauge
7
8 % Material Properties
9 name = 'M42 Steel'; % material name
10 E = 32.E6; % Young's modulus (psi)
11 nu = 0.29; % Poisson's ratio
12 su = 370.E3; % tensile strength use yield or ultimate depending on material (psi)
13 KIC = 15.E3; % fracture toughness (psi sqrt(in))
14 sfatigue = 115.e3; % fatigue strength from Granta for 10^6 cycles
15
16 % Design Constraints
17 a = 0.04; % Assumed crack depth (inches)
18 GF = 2.0; % Assumed Gauge Factor
19
20 % Required Safety Factors
21 XO_req = 4.0; % Static strength safety factor
22 XK_req = 2.0; % Crack growth safety factor
23 XS_req = 1.5; % Fatigue stress safety factor
24 Sensitivity_req = 1.0; % mV/V output at rated torque
25
26 P = M/L;
27 I = (b*h^3)/12;
28 sa = (M*h/2)/I; % max normal stress
29 delta = (P * L^3)/(3*E*I); % deflection
30 K = 1.12*sa*sqrt(pi*a);
31
32 % Stress and deflection analysis
33 fprintf('load point deflection: %.3f in\n', delta); % deflection
34 fprintf('max normal stress: %.2f ksi\n', sa/1000); % max normal stress
35
36 % Calculate safety factors
37 XO = su/sa; % safety factor for strength
38 XK = KIC/K; % safety factor for crack growth
39 XS = sfatigue/sa; % safety factor for fatigue
40
41 % Display safety factor results
42 fprintf('safety factor for strength: %.2f\n', XO);
43 fprintf('safety factor for crack growth: %.2f\n', XK);
44 fprintf('safety factor for fatigue: %.2f\n', XS);
45
46 % Calculate the strain at the gauge
47 Mg = P*(L-c);
48 strainAtGauge = Mg*(h/2)/(E*I);
49 fprintf('strain at gauge: %.0f microstrain\n', strainAtGauge*10^6);
50
51 % Calculate output voltage based on strain gauge sensitivity
52 outputVoltage = 0.5*GF*strainAtGauge*1000;
53 fprintf('output voltage at 600 in-lbf using half bridge: %.2f mV/V\n', outputVoltage);
54
55 % Check if safety factors meet requirements
56 if XO < XO_req
57     fprintf('Warning: Safety factor for strength (%.2f) is below the required (%.2f).\n', XO, XO_req);
58 end
59 % Check if crack growth safety factor meets requirements
60 if XK < XK_req
61     fprintf('Warning: Safety factor for crack growth (%.2f) is below the required (%.2f).\n', XK, XK_req);
62 end
63 % Check if fatigue safety factor meets requirements
64 if XS < XS_req
65     fprintf('Warning: Safety factor for fatigue (%.2f) is below the required (%.2f).\n', XS, XS_req);
66 end
67 % Check if strain gauge output meets sensitivity requirements
68 if outputVoltage < Sensitivity_req
69     fprintf('Warning: Output voltage (%.2f mV/V) is below the required (%.2f mV/V).\n', outputVoltage, Sensitivity_req);
70 end
71 % Final assessment of safety factors and output voltage
72 if XO >= XO_req && XK >= XK_req && XS >= XS_req && outputVoltage >= Sensitivity_req
73     fprintf('All safety factors and output voltage meet the required standards.\n');
74 else
75     fprintf('Some safety factors or output voltage do not meet the required standards.\n');
76 end
```

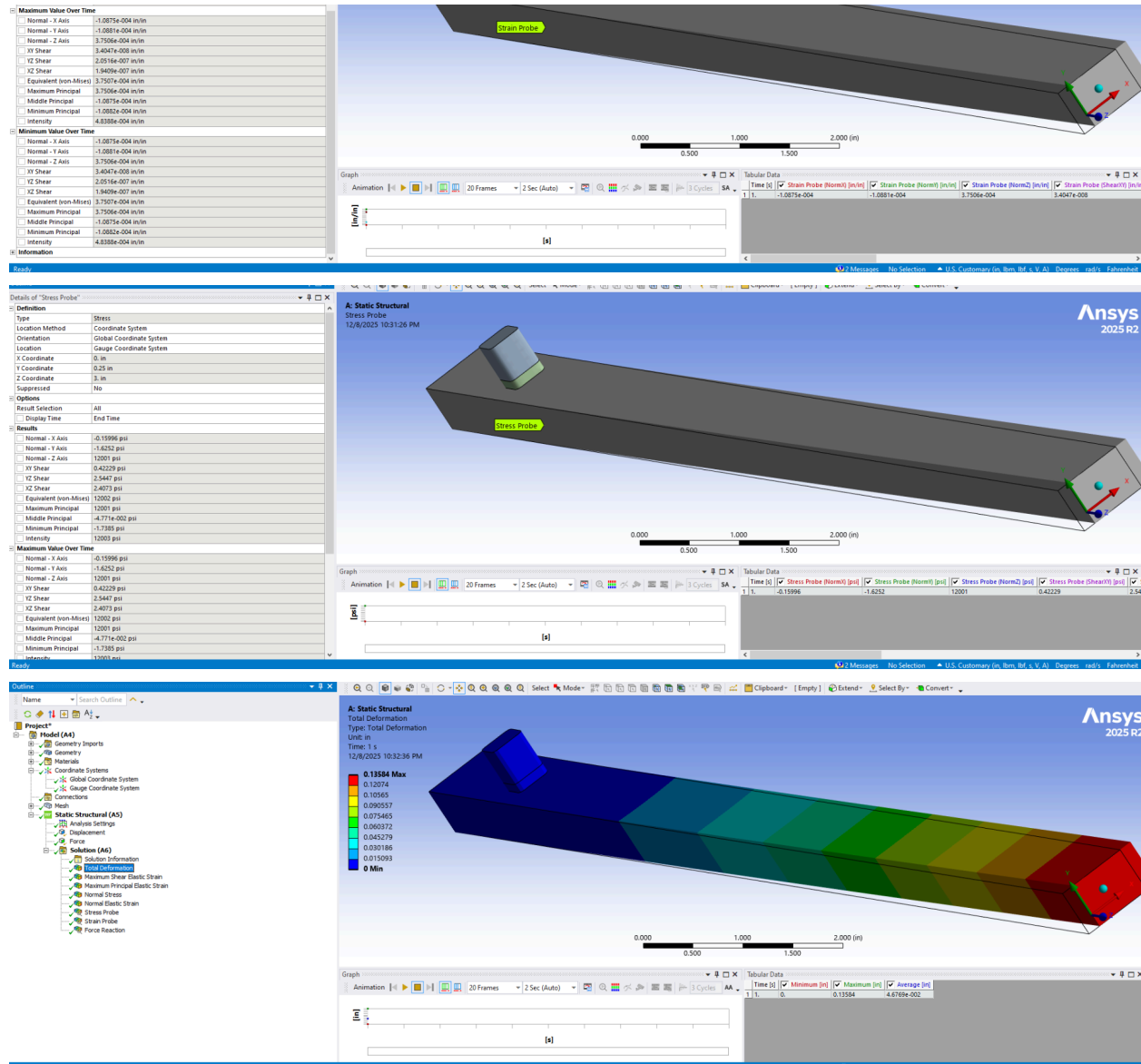
```
>> MAE_3270_Final_HW
load point deflection: 0.091 in
max normal stress: 12.80 ksi
safety factor for strength: 28.91
safety factor for crack growth: 2.95
safety factor for fatigue: 8.98
strain at gauge: 375 microstrain
output voltage at 600 in-lbf using half bridge: 0.38 mV/V
Warning: Output voltage (0.38 mV/V) is below the required (1.00 mV/V).
Some safety factors or output voltage do not meet the required standards.
```

A: Static Structural
 Normal Stress
 Type: Normal Stress(Z Axis)
 Unit: psi
 Global Coordinate System
 Time: 1 s
 12/8/2025 10:25:42 PM

51840 Max
 40081
 28321
 16562
 4803.1
 -6956.1
 -18715
 -30474
 -42234
-53993 Min

0.000 1,000 2,000 (in)

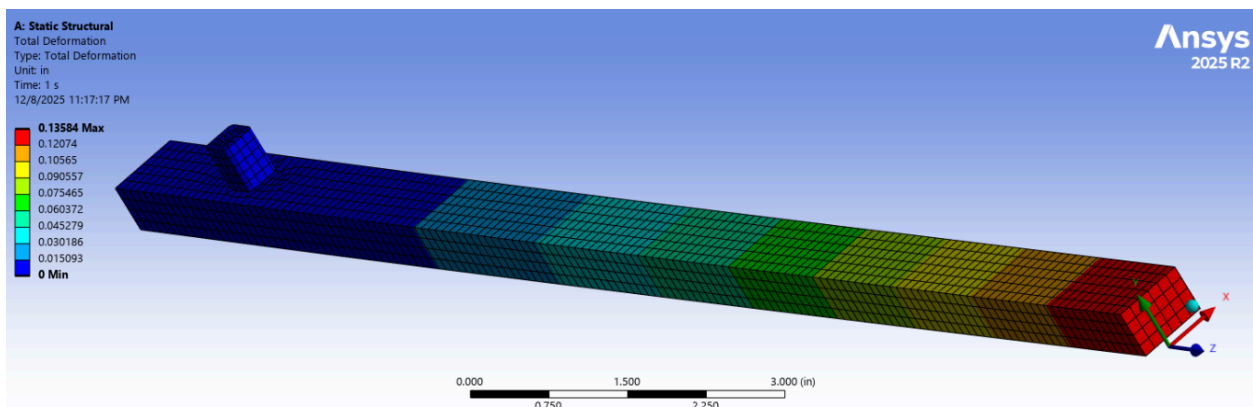
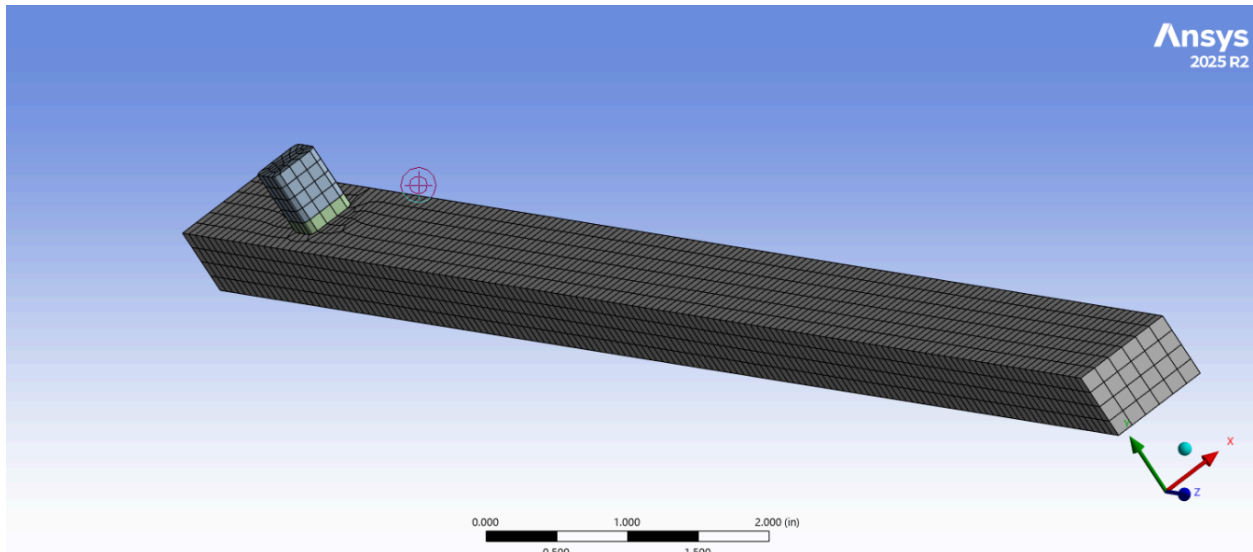




Maximum Normal stress: 51840
 Strain at strain gauge: 3.75×10^{-4}
 Deflection at load point: 0.13584 in

5.1.2 Reflections

1. Beam theory assumes that plane sections remain plane. View the deformed mesh and check if mesh lines that cut across the beam handle remain as straight lines. Do you think that beam theory is reasonably accurate?



The mesh at the tip of the beam handle remains as straight lines. The mesh at the base of the driver deformed because of the fillets. The beam theory is reasonably accurate at regions outside of the drive. So overall, beam theory gives a reasonably accurate picture for the straight, uniform part of the handle, but it's not as reliable right at the transition into the driver body where the geometry changes.

2. How do the FEM and hand-calculated maximum normal stresses compare? If they differ significantly, why?

The FEM maximum normal stress was 51,840 psi, while the hand-calculated value was 12.80 ksi (12,800 psi), so the FEM result is about four times larger. This is expected because the hand calculation is based on simple beam theory, which assumes no fillets and a uniformly distributed

stress over the cross section. On the other hand, the FEM model factors in 3D effects, clamps, and stress concentrations near the fillets. The maximum FEM stress occurs near the fillet, while the hand calculation gives an average bending stress away from these features.

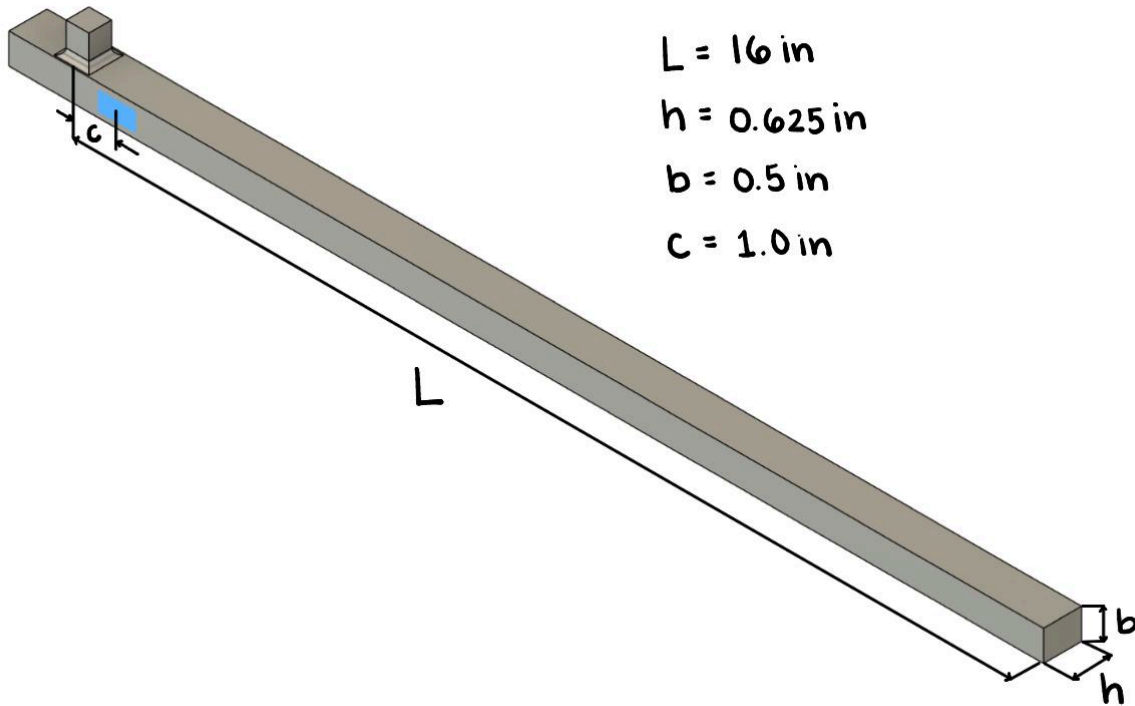
3. How do the FEM and hand-calculated displacements compare? If they differ, why?

The FEM displacement at the load point is 0.13584 in, while the hand-calculated displacement at the load point is 0.091 in. They differ because the FEM model captures flexibility that the hand calculations ignore. In the hand calcs, we assume ideal conditions such as uniform cross-sections and linear elastic behavior. These assumptions often neglect other effects like shear deformation, stress concentrations, and geometric discontinuities. The FEM solution also divides the structure into multiple elements and evaluates stiffness and deformation more accurately along the geometry. The difference arises because FEM provides a more realistic representation of the structure's behavior, while the hand calculations are based on idealized assumptions.

5.2 Your Design, Upload to portfolio

5.2.1 Results

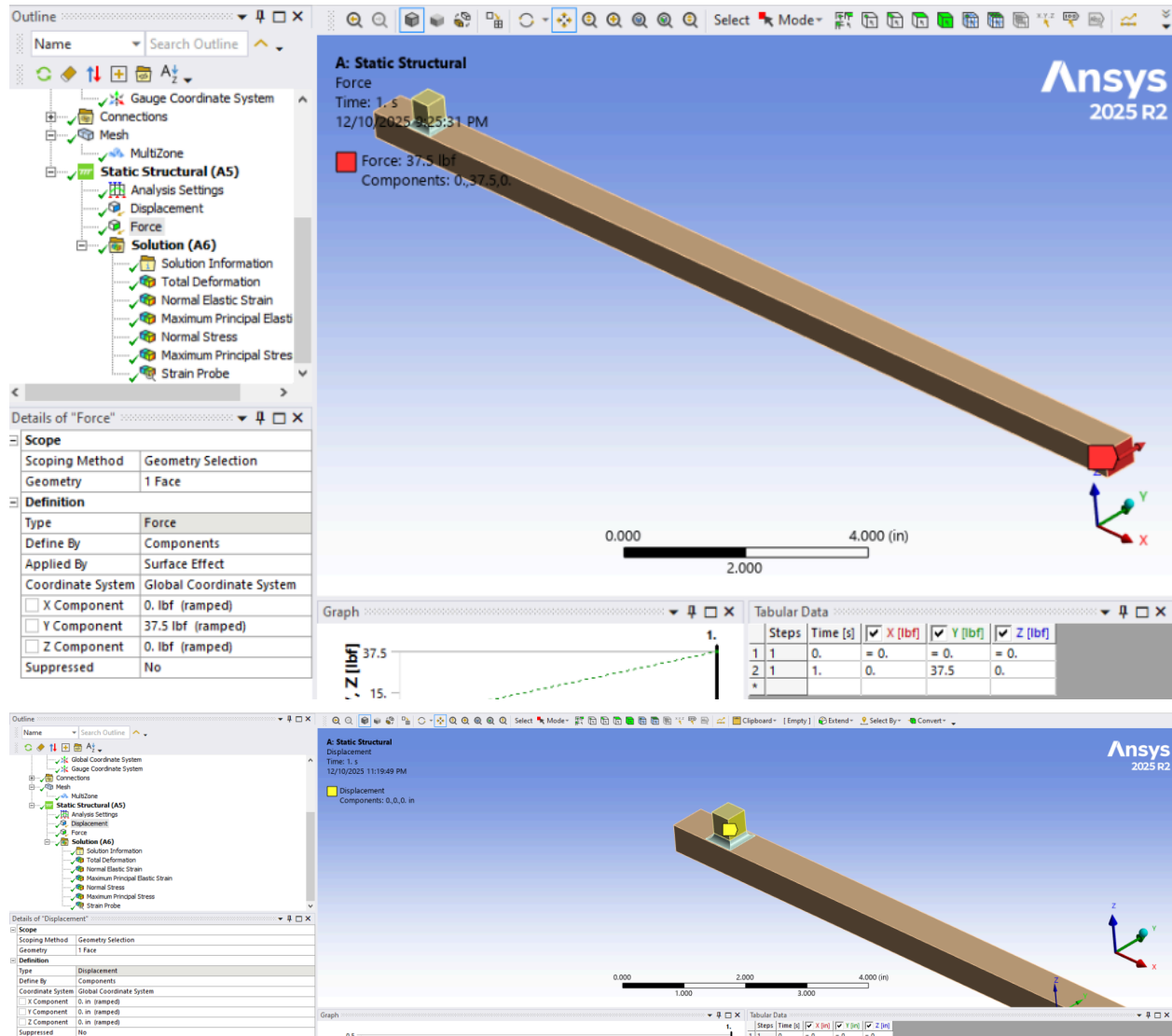
1. Image(s) of CAD model. Must show all key dimensions.



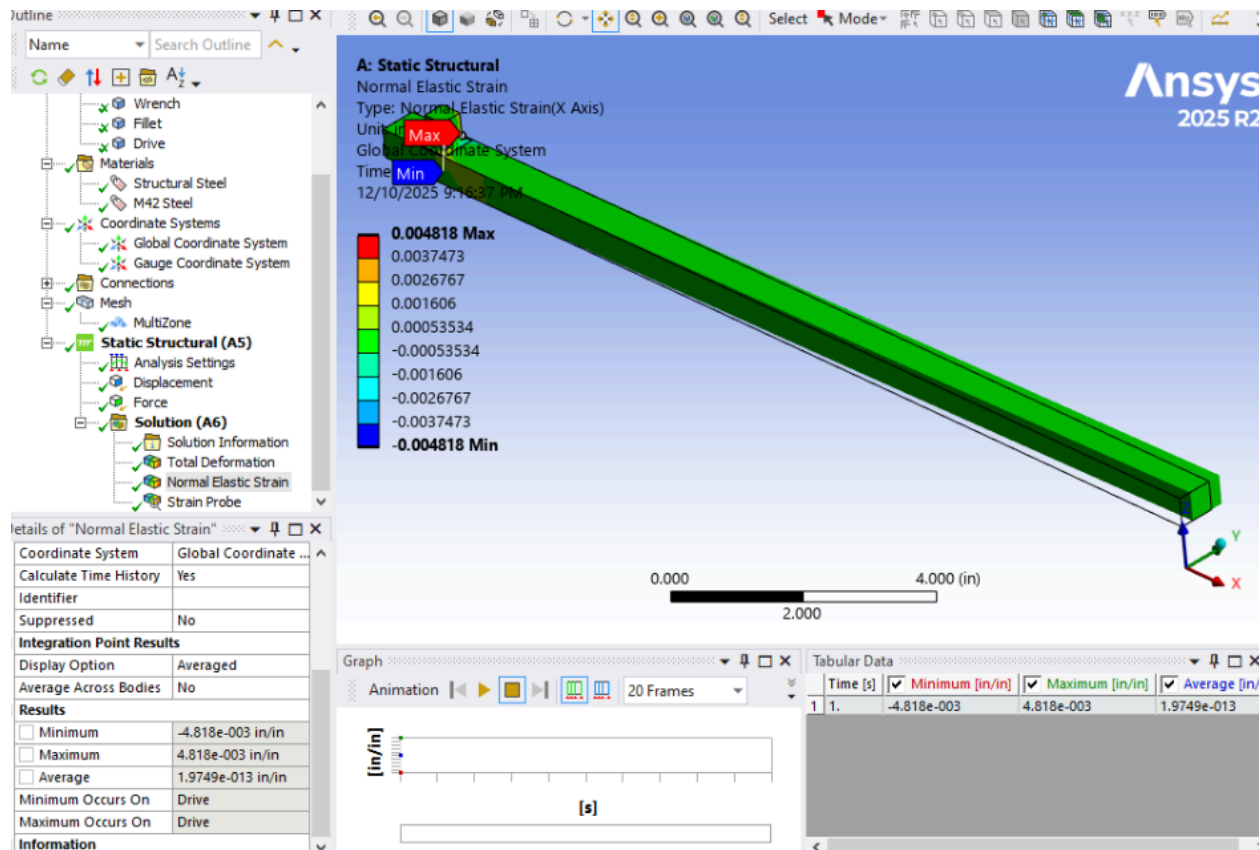
2. Describe material used and its relevant mechanical properties.

```
% Material Properties
name = 'Aluminum 7449'; % material name
E = 10.E6; % Young's modulus (psi)
nu = 0.323; % Poisson's ratio
su = 82.E3; % tensile strength use yield or ultimate depending on material (psi)
KIC = 25.E3; % fracture toughness (psi sqrt(in))
sfatigue = 29.e3; % fatigue strength from Granta for 10^6 cycles
```

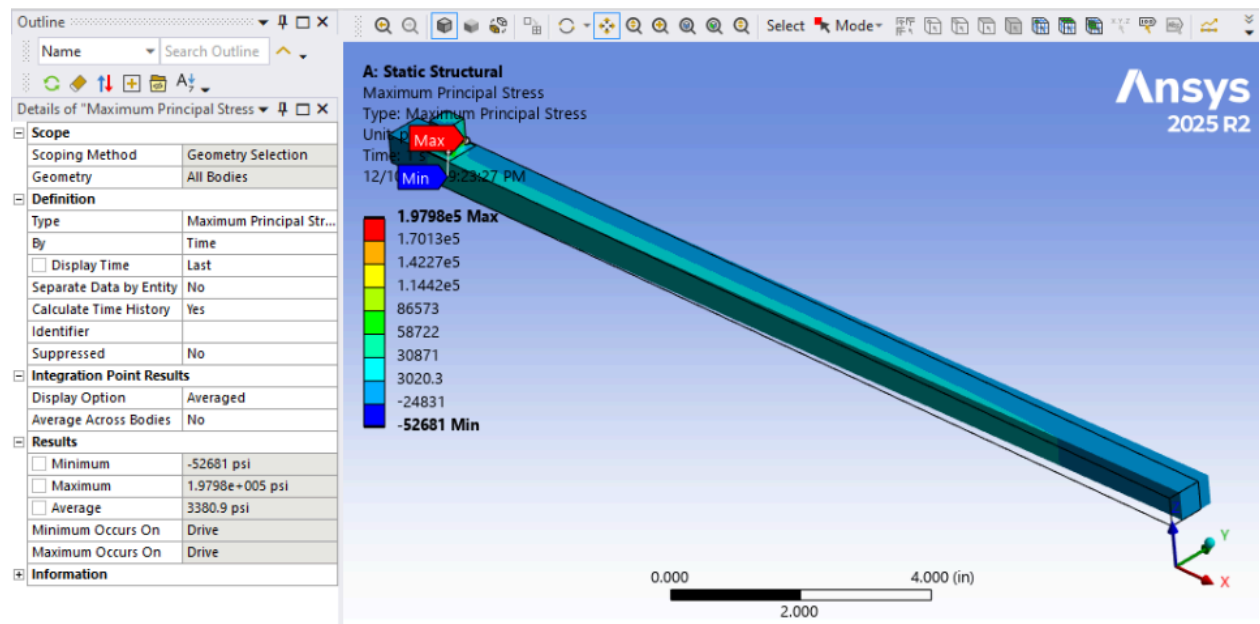
3. Diagram communicating how loads and boundary conditions were applied to your FEM model.



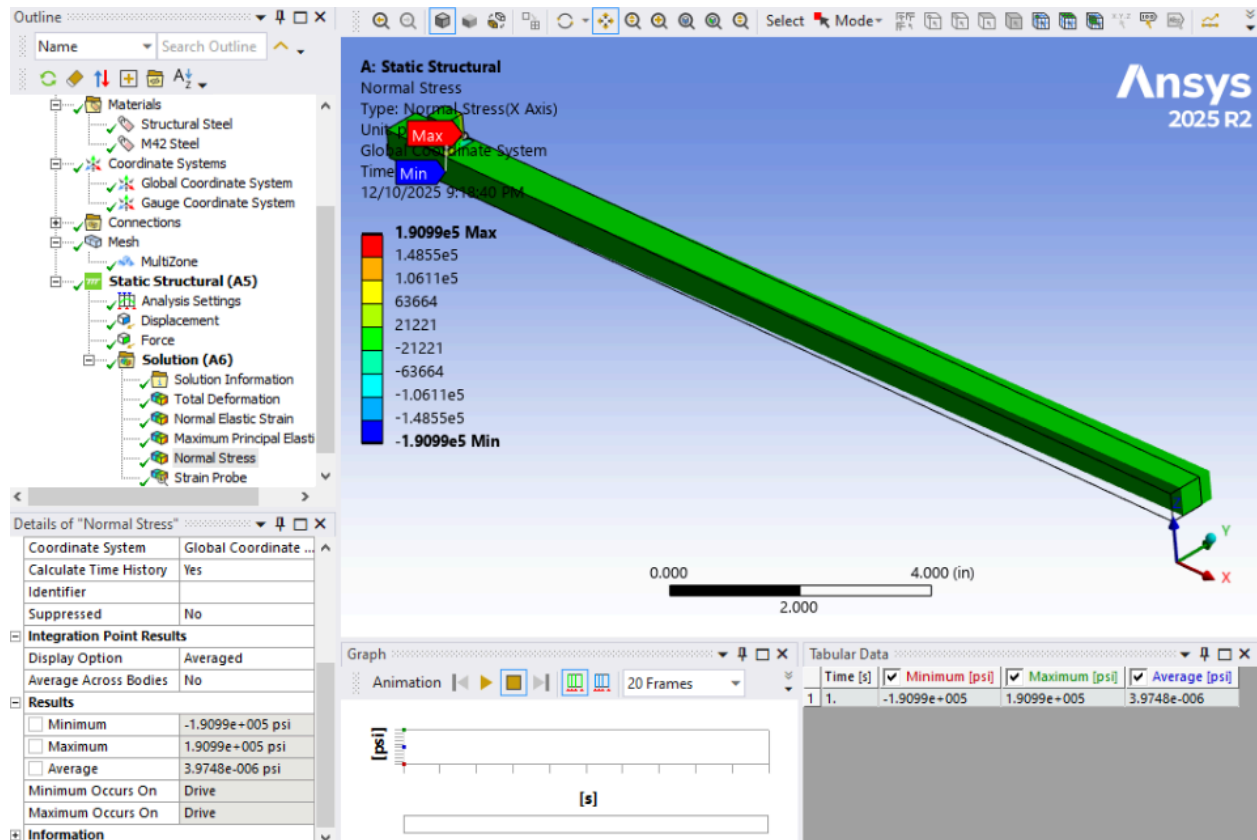
4. Normal strain contours (in the strain gauge direction) from FEM

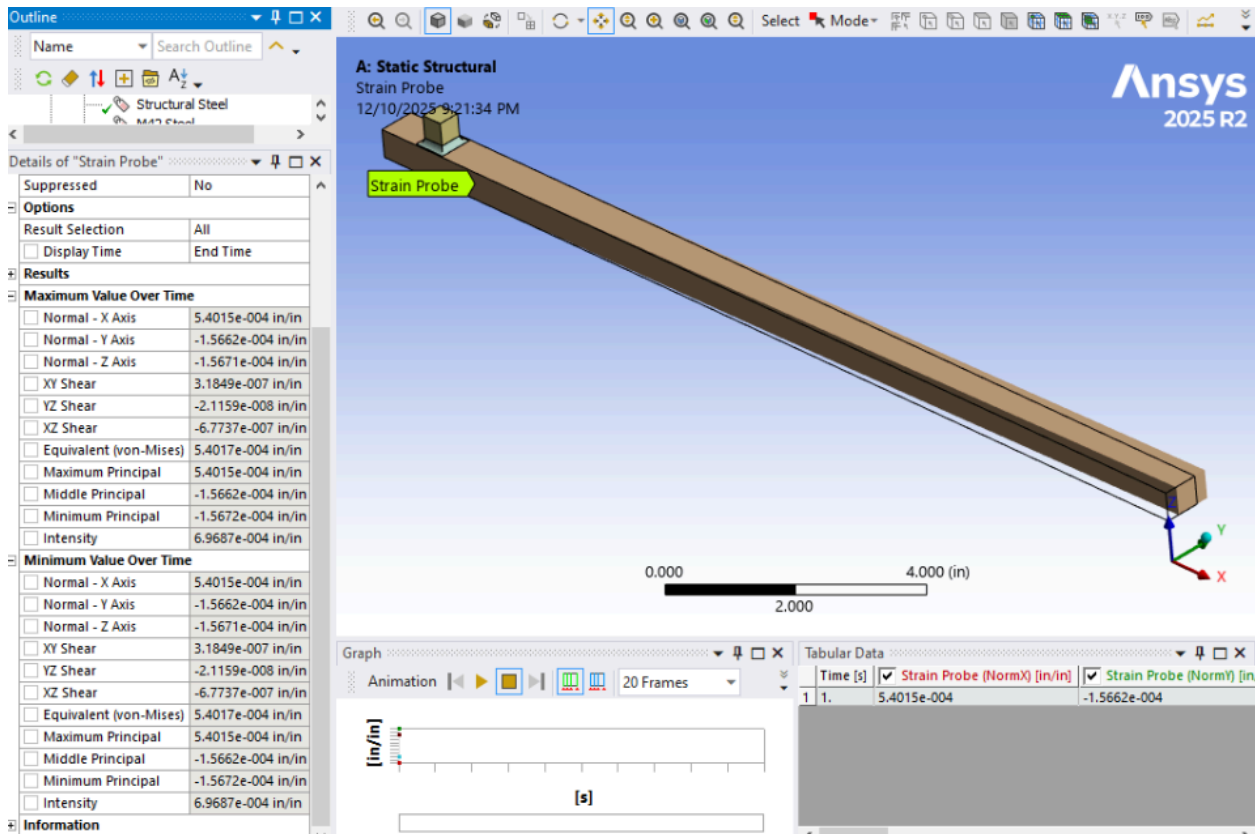
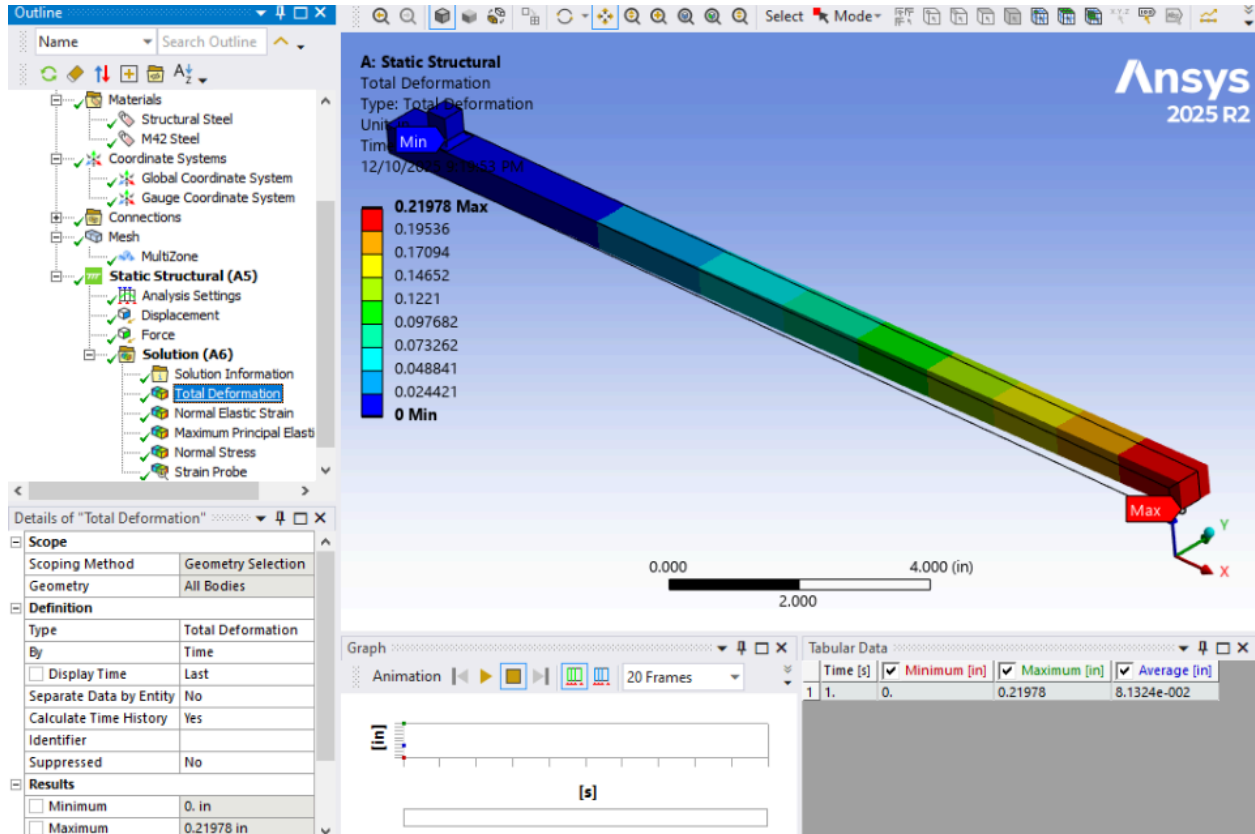


5. Contour plot of maximum principal stress from FEM



6. Summarize results from FEM calculation showing maximum normal stress (anywhere), load point deflection, strains at the strain gauge locations





7. Torque wrench sensitivity in mV/V using strains from the FEM analysis

$$\frac{V_{out}}{V_{in}} = \frac{k}{4} 2\varepsilon, k=2$$

$$1 \frac{mV}{V} = 0.001$$

$$\varepsilon = 5.4015 \times 10^{-4}$$

$$\frac{V_{out}}{V_{in}} = \frac{k}{4} 2\varepsilon = 2.70075 \times 10^{-4}$$

Sensitivity: 0.270 mV/V

8. Strain gauge selected (give type and dimensions). Note that design must physically have enough space to bond the gauges.

SGD-3/350-LY11 (3mm Grid Length, 1.5 mm Grid Width, 120 $\square$ Resistance)